MODULE 38 · DOMAIN 4 - RESILIENCE, MIGRATION AND CONTINUITY

DR Testing and Chaos Engineering

Assess resilience against stated objectives, automate recovery steps, and prove them by injecting real failures under a steady-state hypothesis with stop conditions.

DOMAIN 4 of 5 EXAM WEIGHT 15% MODULE 38 of 52

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A recovery plan nobody has tested is a **hypothesis**. Testing turns it into evidence.

AWS Resilience Hub, next generation

ELEMENT	PURPOSE
Resilience policy	SLOs, RTO and RPO, multi-AZ and multi-Region requirements
System, journeys, services	Model of the application and what users depend on
Dependency discovery	Finds hidden calls from VPC DNS query logs
Failure mode assessment	AI agents find gaps against policy and best practice
Recommended resilience tests	FIS experiments with pass or fail results

ON THE EXAM

Assess against stated RTO and RPO, across accounts, with fixes and tests: Resilience Hub.

Recovery procedures as code

- **Systems Manager Automation runbooks** encode each recovery step.
- **Predefined runbooks** exist for common actions, such as restoring a database.
- **Approval steps** pause for a person where judgement is needed.
- **Run from the recovery Region**, not the one that failed.
- **Every run is logged**, which doubles as evidence.

> ON THE EXAM
>
> Procedures that run the same way every time, with an audit trail: Automation runbooks.

A chaos experiment

01
Steady state
Define normal in business metrics

02
Hypothesis
Steady state holds when X fails

03
Inject
Run the fault with FIS

04
Observe
Stop if alarms breach

05
Learn
Fix gaps, rerun, automate

ON THE EXAM

The hypothesis is written in customer-visible metrics, not CPU or instance counts.

AWS Fault Injection Service

ELEMENT	ROLE
Experiment template	Actions, targets, duration, role
Actions	Stop instances, stress CPU, add latency, disrupt network
Targets	Resources selected by tag, ID or filter, with a percentage
Stop conditions	CloudWatch alarms that halt the experiment
Scenario library	Ready-made AZ power, cross-AZ, cross-Region, EKS, EBS scenarios
Multi-account	Run one experiment across several accounts

ON THE EXAM

Stop conditions are what make chaos safe in production.

Measuring recovery time

- **Start the clock** when the fault is injected, not when someone notices.
- **Route 53 health checks** record when an endpoint failed and when traffic moved.
- **CloudWatch alarms and synthetic canaries** mark when customers were served again.
- **Region switch reports** record each step's timing and the RTO achieved.
- **Compare to the policy**; a miss is a finding to fix, not a footnote.

> ON THE EXAM
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> Measured recovery time is the time until customers are served again, not until servers start.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Assess against RTO and RPO across many apps	Resilience Hub policies and assessments
Hidden dependencies nobody documented	Resilience Hub dependency discovery
Repeatable recovery steps with audit trail	SSM Automation runbooks
Prove it survives an AZ power loss	FIS AZ power interruption scenario
Stop if customers are affected	FIS stop conditions on alarms
What should the hypothesis measure?	Customer-facing steady-state metrics

ON THE EXAM

Assess, automate, then inject and measure, with a safety stop.

> Meridian Freight's 40 critical applications across 25 accounts each have a stated RTO and RPO. Nobody knows which applications can actually meet them.
>
> The resilience team wants continuous assessment against those objectives, with specific gaps and fixes, managed from one account.

What should the solutions architect recommend?

A. A one-off Well-Architected review of each application

B. AWS Resilience Hub with resilience policies, failure mode assessments, and an Organizations delegated administrator

C. AWS Config rules checking Multi-AZ settings

D. A spreadsheet in which owners self-certify

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A One-off review

Not continuous, and not tied to each RTO and RPO.

B Resilience Hub

Policies per objective, ongoing assessment, gaps and fixes, central.

C Config rules

Check settings, not whether objectives can be met.

D Self-certification

Opinion, not assessment.

THE SIGNAL

Objectives stated, gaps unknown, many accounts: Resilience Hub.

> Meridian Freight's tracking service is designed to survive the loss of an Availability Zone, but it has never been tested. The team wants to test it in production during business hours.
>
> If the order success rate falls below 99.5%, the test must stop immediately.

What should the solutions architect implement?

A Manually stop the instances in one AZ and watch dashboards

B Delete the subnets in one AZ

C Run a Resilience Hub assessment instead of a test

D An FIS experiment from the AZ power interruption scenario, with a stop condition on an order success rate alarm

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Manual stop

Partial simulation, and no automatic stop.

B Delete subnets

Destructive and irreversible.

C Assessment only

Analysis is not a test.

D FIS AZ power scenario

Realistic AZ failure symptoms, halted automatically by the alarm.

THE SIGNAL

Realistic AZ failure, in production, with an automatic stop: FIS scenario plus stop condition.

> Meridian Freight's team is writing the steady-state hypothesis for its first chaos experiment on the booking service.
>
> The hypothesis must show whether customers are affected.

Which steady state should the hypothesis use?

A Average CPU utilisation stays below 70%

B The number of running instances stays constant

C Booking success rate stays above 99.9% and p99 latency below 400 ms

D No errors appear in the application logs

◆ PAUSE AND DECIDE

Which steady state should the hypothesis use?

A CPU

Internal; customers may suffer at low CPU.

B Instance count

Expected to change when a fault is injected.

C Success rate and latency

What customers experience, measured continuously.

D No log errors

Any real failure produces some errors; too brittle.

THE SIGNAL

Steady state is what customers experience: success rate and latency.

> Meridian Freight's auditors want quarterly evidence of how long the platform takes to recover when the primary Region loses connectivity to the secondary, measured from the moment of failure.
>
> The test should exercise the real failover path, including DNS.

What should the solutions architect design?

A. Run the FIS cross-Region connectivity scenario, and measure from injection to recovery using Route 53 health check history and canary alarms

B. Ask teams to estimate recovery time in a tabletop exercise

C. Read the Aurora replica lag metric

D. Restore last night's backup in a test account and time it

◆ PAUSE AND DECIDE

What should the solutions architect design?

A FIS plus measured timeline

Real fault, real failover path, timestamps from injection to service.

B Tabletop estimate

Useful practice, but not measurement.

C Replica lag

Indicates RPO, not recovery time.

D Backup restore

Tests a different path from the live failover.

THE SIGNAL

Measured recovery time: inject a real fault, timestamp until customers are served.

> Meridian Freight's DR procedure for its claims system has 14 manual steps, including restoring an RDS snapshot, updating a Parameter Store value and restarting ECS services. Steps are often missed under pressure.
>
> The company wants the steps to run identically every time, with a record of each run.

What should the solutions architect implement?

A. A more detailed wiki page with screenshots

B. A Systems Manager Automation runbook encoding the steps, run from the recovery Region

C. A cron job on a server in the primary Region

D. Assign one engineer to own the procedure

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Better wiki

Still manual, still missed under pressure.

B Automation runbook

Same steps every time, logged, independent of the failed Region.

C Cron in primary

Lives in the Region that has failed.

D One owner

A single point of failure in people.

THE SIGNAL

Repeatable recovery steps with a record: an Automation runbook.

> Meridian Freight's checkout path spans 12 accounts: web, payments, inventory and shared networking. The resilience team wants one experiment that injects latency into several of them at once.
>
> Each account's owners must keep control over what can be targeted in their account.

What should the solutions architect use?

A. Twelve separate experiments started at the same moment

B. A Lambda function with cross-account admin roles that stops resources

C. A Resilience Hub assessment of each account

D. An FIS multi-account experiment from an orchestrator account, with target account configurations and roles scoped by each account

◆ PAUSE AND DECIDE

What should the solutions architect use?

A Twelve experiments

Hard to coordinate, and stop conditions do not span them.

B Custom Lambda

Broad admin roles and no built-in safety.

C Assessments

Analysis, not an experiment.

D FIS multi-account

One experiment, one set of stop conditions, owner-scoped roles.

THE SIGNAL

One fault across many accounts, with owner control: FIS multi-account experiments.

What to carry out of this module

- **Resilience Hub**: policies, dependency discovery, AI assessments, recommended tests.
- **Automation runbooks**: recovery steps as code, logged, from the healthy Region.
- **Hypothesis in customer metrics**; stop conditions on the same alarms.
- **FIS scenarios** for AZ power, cross-AZ and cross-Region failures.
- **Measure recovery** from injection until customers are served again.

> ON THE EXAM
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> Assess, automate, inject, measure, and repeat.